Plots and tables

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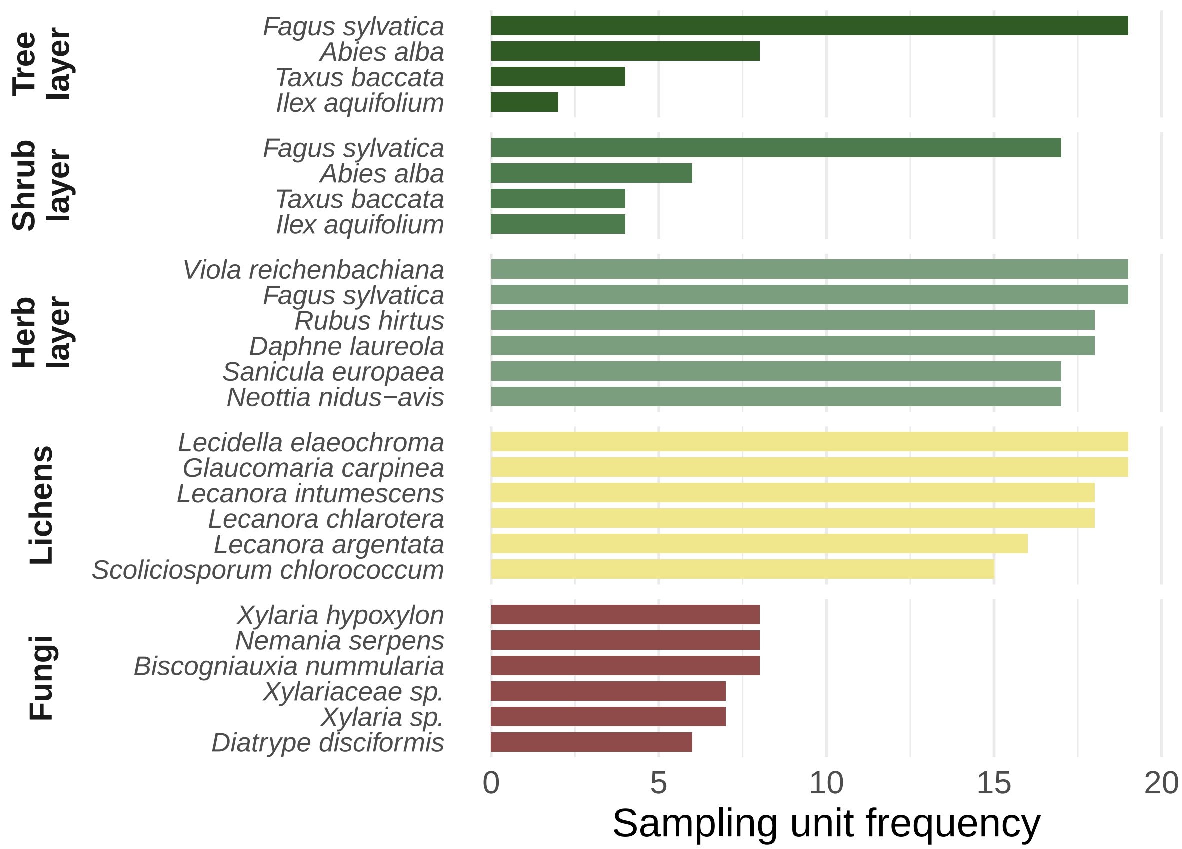
```r
# SCRIPT FOR DATA NOTE GRAPHS
# libraries ----
library(openxlsx)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become error
```

```r
library(patchwork)

# ggplot theme ----

my_theme <- function(...){
  theme_minimal() +
    theme(axis.title.y = element_text(size = 5),
          axis.title.x = element_text(size = 15),
          axis.text.y = element_text(face = "italic", size = 10),
          axis.text.x = element_text(size = 12),
          legend.position = "none", plot.title.position = "panel",
          title = element_text(size = 17),
          strip.text.y = element_text(size = 12, face = "bold"),
          strip.placement = "outside",
          panel.grid.major.y = element_blank(),
          ...)
}


# DATA IMPORT ----
sheets <- loadWorkbook('data/Data2024.xlsx') |>
  names() |>
  set_names()

data <- map(sheets, \(x) readWorkbook('data/Data2024.xlsx', sheet = x))
```

```r
# "MOST FREQUENT SPECIES" plot ----

get_frequency <- function(x, cutoff = 0){
  x |>
    pivot_longer(!plotID) |>
    filter_out(value == 0) |>
    count(name) |>
    arrange(desc(n)) |>
    filter(n > cutoff) |>
    mutate(name = str_replace(name, "(?<=\\w)\\.(?=\\w)", " "))
}

## Plants data
# Tree layer

tree_freq <- data$GS_2024_trees |>
  get_frequency(cutoff = 1)

# Shrubs

shrub_freq <- data$GS_2024_shrubs |>
  get_frequency(cutoff = 1)

# Herbs

herb_freq <- data$GS_2024_herbs |>
  get_frequency(cutoff = 16)

## Lichens data

lichen_freq <- data$GS_2024_lichens |>
  get_frequency(cutoff = 14)

## Fungi data

fungi_freq <- data$GS_2024_fungi |>
  get_frequency(cutoff = 5)


## Dataset binding
freq_all <- bind_rows(tree_freq, shrub_freq, herb_freq, lichen_freq, fungi_freq, .id = "group") |>
  mutate(group = recode_values(group,
    "1" ~ "Tree\nlayer",
    "2" ~ "Shrub\nlayer",
    "3" ~ "Herb\nlayer",
    "4" ~ "Lichens",
    "5" ~ "Fungi")) |>
  mutate(group = as_factor(group))

# "Most frequent species" Graph
gg_spp_freq <- freq_all |>
  ggplot(aes(x = fct_reorder(name, n), y = n, fill = group)) +
  geom_bar(stat = 'identity', width = 0.75) +
```

```r
  scale_fill_manual(values = c(
    "Tree\nlayer" = "#315b25",
    "Shrub\nlayer" = "#4d7b4d",
    "Herb\nlayer" = "#7B9E7E",
    "Lichens" = "#F0E68C",
    "Fungi" = "#8F4A4A")) +
  coord_flip() +
  labs(y = "Sampling unit frequency", x = "") +
  ylim(c(0, 20)) +
  facet_grid(group ~ ., scales = "free", space = "free",
             switch = "both") +
  my_theme()

gg_spp_freq
```




```r
# Save graph - EPS format
ggsave("Fig3.eps",
       plot = gg_spp_freq,
       width = 119,
       height = 73,
       scale = 1.64,
       units = "mm",
       bg = "white")

# Save graph - PNG format
```

```r
ggsave(
  filename = "most_frequent_species.png",
  plot = gg_spp_freq,
  width = 119,
  height = 73,
  scale = 1.64,
  units = "mm",
  bg = "white")

# "FOREST STRUCTURE" plot ----
forest_str <- data$GS_2024_structure |>
  separate_wider_position(plotID, c(site = 2, plot = 1)) %>%
  mutate(site = recode_values(site,
                              "PR" ~ "Prati di Tivo",
                              "IN" ~ "Incodaro",
                              "VE" ~ "Venaquaro"))

# Number of records per each tree species in each study site
forest_str |>
  count(site, treesp) |>
  pivot_wider(names_from = site, values_from = n) |>
  arrange(treesp)
```

```
## # A tibble: 5 x 4
##   treesp              Incodaro `Prati di Tivo` Venaquaro
##   <chr>                  <int>           <int>     <int>
## 1 Abies alba                87              NA        NA
## 2 Acer pseudoplatanus        1              NA        NA
## 3 Fagus sylvatica          382             189       314
## 4 Ilex aquifolium           NA              NA         2
## 5 Taxus baccata             NA               8         1
```

```r
# Creation of a common legend among study site
species_str <- forest_str |>
  distinct(treesp) |>
  arrange(treesp) |>
  pull()

species_str
```

```
## [1] "Abies alba"          "Acer pseudoplatanus" "Fagus sylvatica"
## [4] "Ilex aquifolium"     "Taxus baccata"
```

```r
col_leg <- scale_fill_manual(
  name   = "Species",
  limits = species_str,
  drop   = FALSE,
  values = c(
    "Abies alba" = "#0072B2",
    "Acer pseudoplatanus" = "#E69F00",
    "Fagus sylvatica" = "#56B4E9",
    "Ilex aquifolium" = "#000000",
```

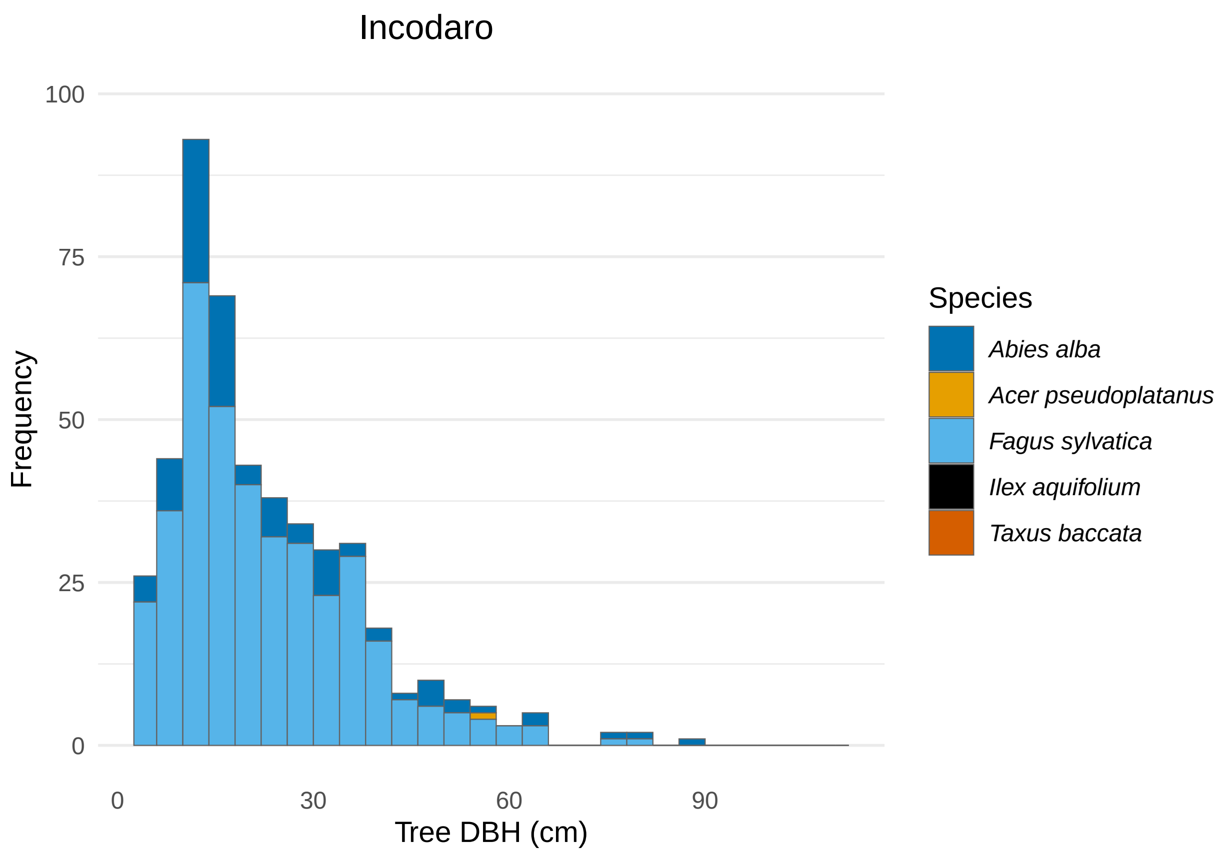
```r
    "Taxus baccata" =  "#D55E00"
  )
)

# Ranges, means, medians, quantiles and histograms of DBH in...

my_summary <- function(x, var){
  x |>
    summarise(
      min    = min({{var}}, na.rm = T),
      max    = max({{var}}, na.rm = T),
      mean   = mean({{var}}, na.rm = T),
      median = median({{var}}, na.rm = T),
      q25    = quantile({{var}}, probs = 0.25, na.rm = T),
      q75    = quantile({{var}}, probs = 0.75, na.rm = T)
    )
}

hist_str <- function(x, where, what, xlim, ylim, binwidth, xlab){
  x |>
    filter(site == where) |>
    ggplot(aes({{what}}, fill = treesp)) +
    geom_histogram(color = "#636363", binwidth = binwidth, linewidth = 0.2,
                   show.legend = T) +
    col_leg +
    labs(x = xlab, y = "Frequency", fill = "Species", title = where)  +
    scale_x_continuous(limits = xlim, oob = scales::squish) +
    ylim(ylim) +
    theme_minimal() +
    theme(plot.title = element_text(hjust = 0.4),
          legend.text = element_text(face = "italic"),
          panel.grid.major.x = element_blank(),
          panel.grid.minor.x = element_blank())
}

# ...entire dataset

forest_str |>
  my_summary(treedb)
```

```
## # A tibble: 1 x 6
##     min   max  mean median   q25   q75
##   <dbl> <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1   2.5   112  25.2     20    13  34.2
```

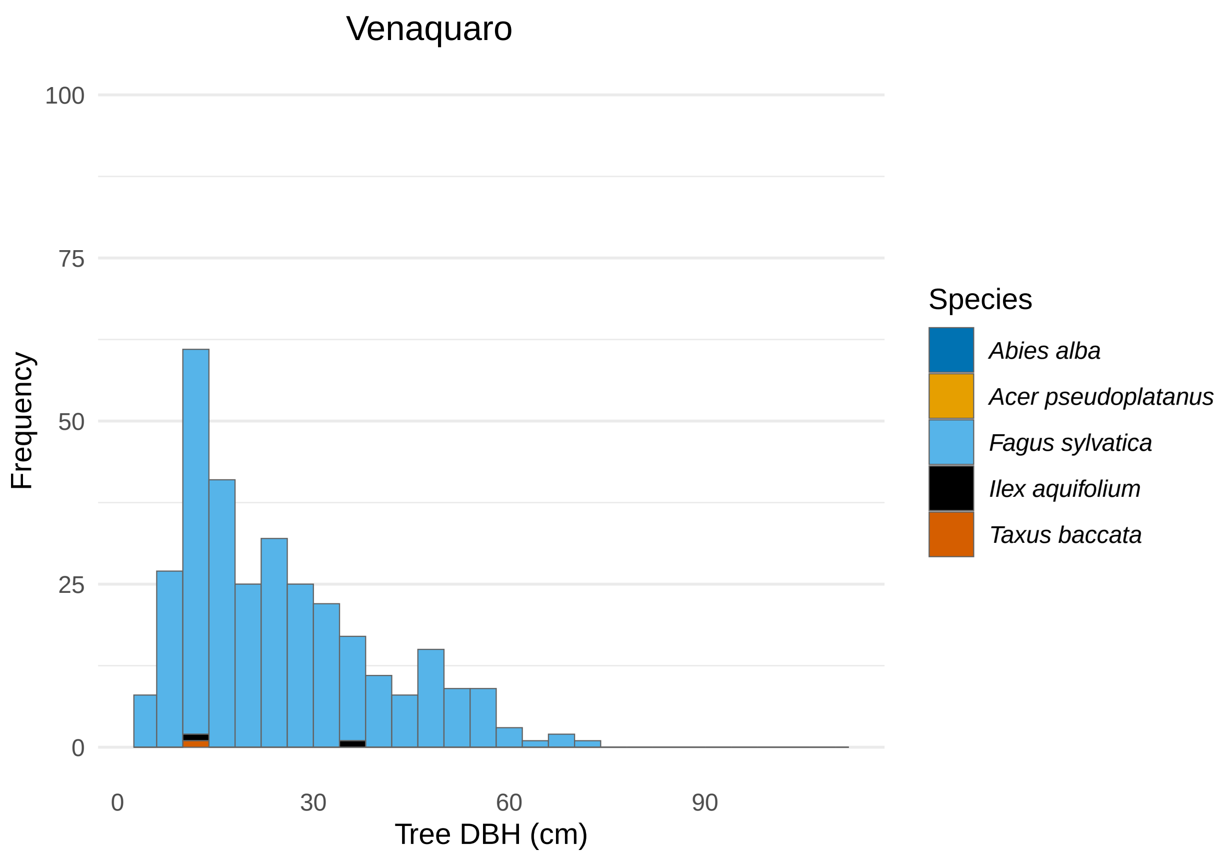
```r
xlim_db <- range(forest_str$treedb)
ylim_db <- c(0, 100)

# ...by site

forest_str |>
  group_by(site) |>
  my_summary(treedb)
```

```
## # A tibble: 3 x 7
##   site           min   max  mean median   q25   q75
##   <chr>        <dbl> <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1 Incodaro       3      88  23.1     19    12    31
## 2 Prati di Tivo  2.5   112  30.3     23    13    48
## 3 Venaquaro      3      73  25.3     22    14    34
```

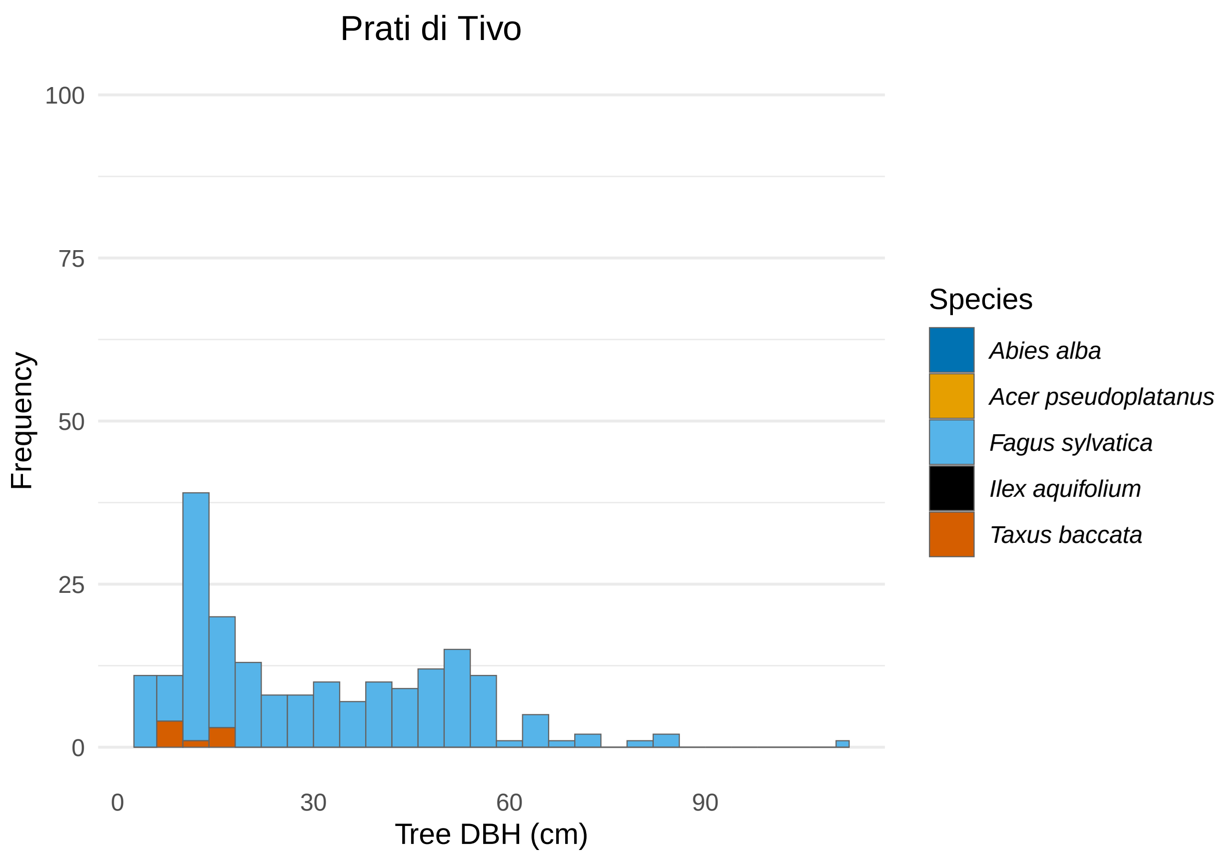
```
# ... Incodaro study site

hist_IN <- forest_str |>
  hist_str("Incodaro", treedb, xlim_db, ylim_db, 4, "Tree DBH (cm)")

hist_IN
```




```
# ... Venaquaro study site

hist_VE <- forest_str |>
  hist_str("Venaquaro", treedb, xlim_db, ylim_db, 4, "Tree DBH (cm)")

hist_VE
```




```r
# ... Prati di Tivo study site

hist_PR <- forest_str |>
  hist_str("Prati di Tivo", treedb, xlim_db, ylim_db, 4, "Tree DBH (cm)")

hist_PR
```




```
# Ranges, means, medians, quantiles and histograms of height measured in...
# ... entire dataset
forest_str |>
  my_summary(treeht)
```

```
## # A tibble: 1 x 6
##     min   max  mean median   q25   q75
##   <dbl> <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1     2    42  19.4   20.2    12    27
```

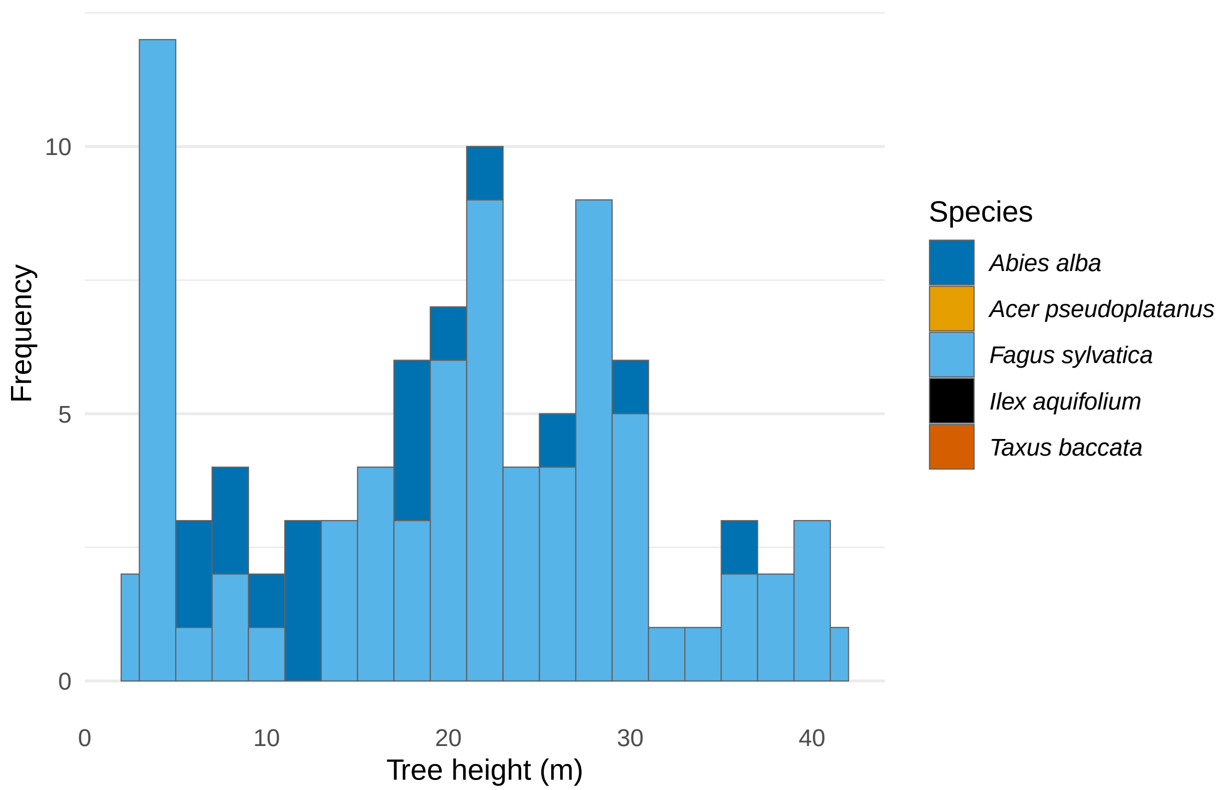
```
xlim_ht <- range(forest_str$treeht, na.rm = T)
ylim_ht <- c(0,13)

# ... by site

forest_str |>
  group_by(site) |>
  my_summary(treeht)
```

```
## # A tibble: 3 x 7
##   site             min   max  mean median   q25   q75
##   <chr>          <dbl> <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1 Incodaro         2    42    19.4   20.8   8    28
## 2 Prati di Tivo    2.5  38    20.0   21.8  13.4  27
## 3 Venaquaro        2.5  35.1  19.0   19.7  13.5  24.2
```

```r
# ... Incodaro study site

hist_ht_IN <- forest_str |>
  hist_str("Incodaro", treeht, xlim_ht, ylim_ht, 2, "Tree height (m)")+
  ggtitle(NULL)

hist_ht_IN
```

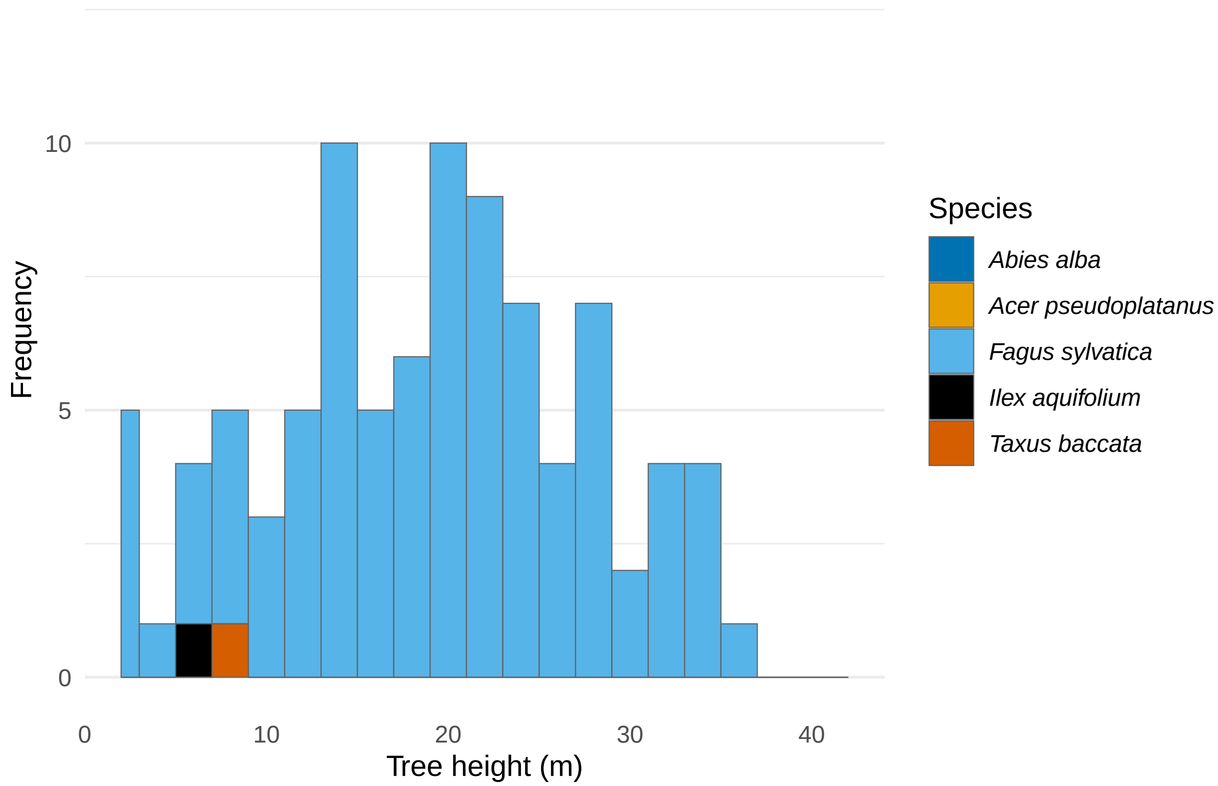
```
## Warning: Removed 374 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_bar()`).
```



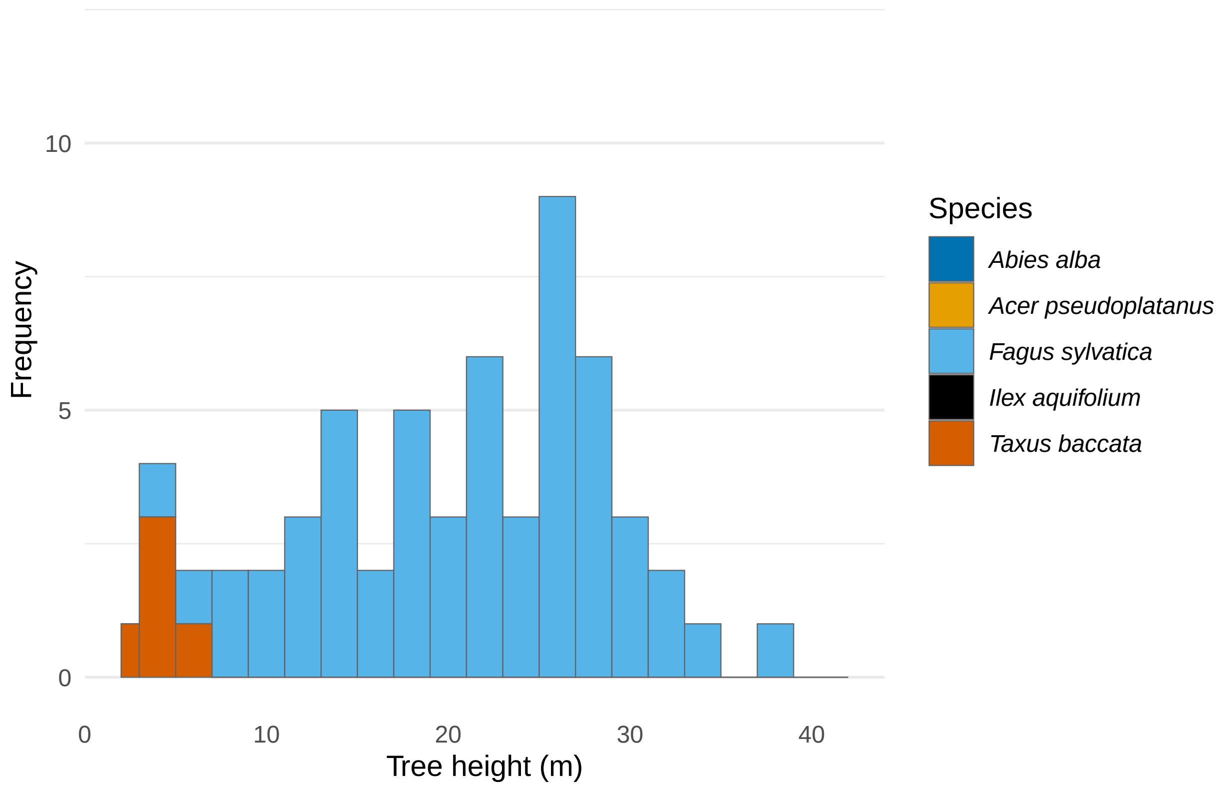

```r
# ... Venaquaro study site
hist_ht_VE <- forest_str |>
  hist_str("Venaquaro", treeht, xlim_ht, ylim_ht, 2, "Tree height (m)")+
  ggtitle(NULL)

hist_ht_VE
```

```
## Warning: Removed 225 rows containing non-finite outside the scale range
## (`stat_bin()`).
```



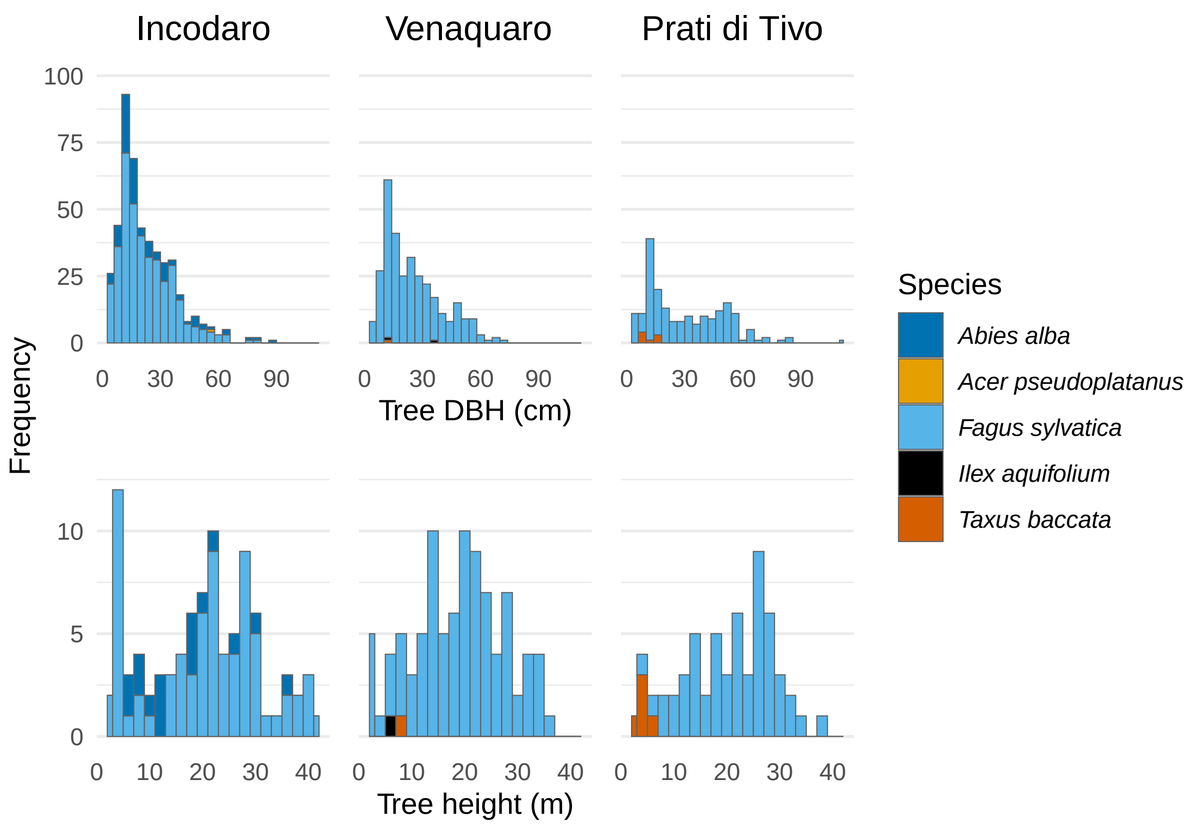

```
# ... Prati di Tivo study site
hist_ht_PR <- forest_str |>
  hist_str("Prati di Tivo", treeht, xlim_ht, ylim_ht, 2, "Tree height (m)")+
  ggtitle(NULL)

hist_ht_PR
```

```
## Warning: Removed 137 rows containing non-finite outside the scale range
## (‘stat_bin()‘).
```




```r
# Combined DBH and heights histograms of the three sites
combi_plots <- wrap_plots(hist_IN, hist_VE, hist_PR,
                          hist_ht_IN, hist_ht_VE, hist_ht_PR,
                          ncol = 3, nrow = 2,
                          guides = "collect", axes = "collect")

combi_plots
```

```
## Warning: Removed 374 rows containing non-finite outside the scale range (‘stat_bin()‘).
## Removed 1 row containing missing values or values outside the scale range
## (‘geom_bar()‘).

## Warning: Removed 225 rows containing non-finite outside the scale range
## (‘stat_bin()‘).

## Warning: Removed 137 rows containing non-finite outside the scale range
## (‘stat_bin()‘).
```




```
# Save graph - EPS format
ggsave("Fig4.eps",
       combi_plots,
       width = 119,
       height = 73,
       scale = 1.7,
       units = "mm",
       bg = "white")
```

```
## Warning: Removed 374 rows containing non-finite outside the scale range
## (`stat_bin()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_bar()`).

## Warning: Removed 225 rows containing non-finite outside the scale range
## (`stat_bin()`).

## Warning: Removed 137 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

```
# Save graph - PNG format
ggsave(
  filename = "forest_structure.png",
```

```r
  combi_plots,
  width = 119,
  height = 73,
  scale = 1.7,
  units = "mm",
  bg = "white")
```

```
## Warning: Removed 374 rows containing non-finite outside the scale range
## (`stat_bin()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_bar()`).

## Warning: Removed 225 rows containing non-finite outside the scale range
## (`stat_bin()`).

## Warning: Removed 137 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

```r
# DEADWOOD table ----

# entire dataset
data$GS_2024_deadwood |>
  pivot_longer(diam1:height) |>
  group_by(name) |>
  my_summary(value)
```

```
## # A tibble: 4 x 7
##   name     min   max  mean median   q25   q75
##   <chr>  <dbl> <dbl> <dbl>  <dbl> <dbl> <dbl>
## 1 diam1      8   110  22.5     17    12    28
## 2 diam2      8    99  31.7     29    19    40
## 3 height     4   120  23.2     16    12    25
## 4 length    10  1800 199.     110    91   190
```

```r
#by site
data$GS_2024_deadwood |>
  pivot_longer(diam1:height) |>
  mutate(siteID = str_sub(plotID, 1, 2)) |>
  select(!c(plotID, lynID)) |>
  group_by(name, siteID) |>
  my_summary(value)
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by name and siteID.
## i Output is grouped by name.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(name, siteID))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```
## # A tibble: 12 x 8
## # Groups:   name [4]
##    name   siteID   min   max  mean median   q25   q75
##    <chr>  <chr>  <dbl> <dbl> <dbl>  <dbl> <dbl> <dbl>
##  1 diam1  IN       8     110  19.0   15    11    22
##  2 diam1  PR       8     110  30.6   20    12    44.8
##  3 diam1  VE       8      61  26.3   22    13    36.2
##  4 diam2  IN      11.5    64  28.9   30    16.2  37
##  5 diam2  PR      11      70  40.2   40    26.5  52.2
##  6 diam2  VE       8      99  29.6   26    17.2  39.8
##  7 height IN       4      73  18.3   15    11    20
##  8 height PR       5     120  30.2   16    11.8  35
##  9 height VE       5     110  23.3   19.5  12.2  26
## 10 length IN      10    1800 210.   110    96.2 190
## 11 length PR      16     640 137.    95    66   150
## 12 length VE      13     700 198    137    70   295
```